The Attribution Blind Spot: Layerwise Trajectory Diagnostics for Source Reliance in Retrieval-Augmented Language Models

Anonymous submission

Abstract

A retrieval-augmented model can agree with a document without relying on it, so output agreement cannot identify the governing source. We address this *attribution blind spot* with three tests—verified exposure, observed choice under conflict, and frozen intervention—and use paired with- and without-context passes to define signed, layer-resolved Latent Trajectory Shift (LTS). In one OLMo system, a shared item cohort separates unresolved exposure from predictable source choice and selective control. The direction frozen on that cohort also transfers without refitting. Beyond OLMo, exact Pythia checkpoints bound exposure, while Llama, Qwen, and Mistral repeat the choice–control pattern. On a shared 110-item diagnostic support, magnitude carries most trajectory-based ranking signal, whereas sign enables selective intervention. The result holds against matched-norm controls, transfers across datasets, and extends to a one-model long-form test. Rather than claim a universal source detector, we specify what internal evidence is required before making a source-reliance claim.

1 Introduction

Retrieval-augmented generation (RAG) (Lewis et al. 2020; Guu et al. 2020; Borgeaud et al. 2022) promises more than an answer that happens to match a document. The retrieved evidence should govern generation when appropriate. The answer alone, however, reveals agreement rather than governance.

The problem is sharpest when retrieval repeats knowledge already stored in the model. Reading the passage and recalling the fact can then produce the same text. We call this non-identifiability the **attribution blind spot**. Membership inference asks whether a document occurred in training (Shi et al. 2024; Duan et al. 2024); faithfulness and citation metrics ask whether a passage supports an answer (Niu et al. 2024; Liu, Zhang, and Liang 2023; Bohnet et al. 2022). Neither question identifies which source governed one shared answer.

We therefore ask three separate questions. **RQ1 (Exposure)** asks what verified training exposure can establish. **RQ2 (Choice)** forces context and memory to disagree, making source preference observable. **RQ3 (Control)** asks whether a direction fitted only on training folds can selectively move that choice and transfer. We use these three tests as an *evidence contract*: each changes what is observable and therefore supports a different source claim (Figure 1).

Prior work studies parts of this chain. It probes conflict activations (Tighidet et al. 2024), measures source allocation when answers agree (Tao et al. 2024), analyzes retrieval-induced representation shifts (Yeh and Li 2026), and steers context–memory preference (Li, Chen, and Tong 2025; Bi et al. 2026). We connect the questions by subtracting hidden states from matched runs with and without context. A training-only projection of this difference gives the signed Latent Trajectory Shift (LTS). Inspired by cognitive reality monitoring (Johnson, Hashtroudi, and Lindsay 1993), the broader Cognitive Reality Monitoring (CRM) framework treats processing traces as evidence about source. LTS makes that hypothesis measurable and intervenable.

We first apply all three tests to OLMo-2-1124-7B-Instruct (Team OLMo et al. 2024). On the same items, exposure is inconclusive at the achieved power, source choices can be ranked, and a frozen LTS direction changes those choices. We then apply that direction to ConflictQA without refitting. Next, we test exposure in exact Pythia checkpoints and test choice and control in Llama, Qwen, and Mistral. Further tests explain the central asymmetry: magnitude carries most predictive signal, whereas sign is needed for selective control. Our contributions are:

1. **A testable definition of source evidence.** We show why context-consistent text cannot identify retrieval use and separate the claims supported by exposure, conflict choice, and intervention.
2. **A common latent measurement.** LTS is fitted without test-fold access and evaluated for held-out prediction, signed control, and transfer. The same-model OLMo chain shows that the three tests are empirically distinct.
3. **Selective, transferable control.** The effect of the frozen intervention replicates across model families and data shifts. On a shared 110-item evaluation, LTS exceeds matched-budget random and answer-gradient alternatives while preserving non-target behavior; a separate one-model study extends the endpoint to long-form answers.

2 Paired Trajectory Differencing

Evidence contract. Let $\mathcal{M}_\theta$ answer query q with context c, and let $S \in \{C, P\}$ denote context or parameters as the governing source. If two causal accounts yield the same observed (q, c, y) but assign different values to S, no statistic of that observation identifies the source. The Supplementary Document gives the formal argument. We therefore change the observable at each stage: paired passes measure state displacement, incompatible answers expose choice, and a frozen intervention measures behavioral response. Membership establishes exposure, not the source of one answer.

Measurement access. CRM records what an evaluator can observe. CRM level 1 measures with/no-context generation distance using BGE-M3 (Chen et al. 2024; Reimers and Gurevych 2019); level 2 aggregates output-head KL divergences; level 3 uses hidden states. These access levels are distinct from the Euclidean ℓ_2-magnitude baseline used below. Access determines the available signal; the experimental stage determines the claim.

Latent Trajectory Shift (LTS). For item i, two matched conditions a and b define a layerwise contrast at the last shared prompt position:

$$\delta_{\ell i}^{a:b} = h_{\ell i}^{(a)} - h_{\ell i}^{(b)}, \quad (1)$$

$$\mu_\ell^{a:b} = \frac{1}{|\mathcal{T}|} \sum_{i \in \mathcal{T}} \delta_{\ell i}^{a:b}, \quad (2)$$

$$u_\ell^{a:b} = \arg \max_{\|u\|_2=1} \sum_{i \in \mathcal{T}} \left[u^\top (\delta_{\ell i}^{a:b} - \mu_\ell^{a:b})\right]^2, \quad (3)$$

$$\mathrm{LTS}_{\ell i}^{a:b} = \langle \delta_{\ell i}^{a:b}, v_\ell^{a:b} \rangle, \qquad \|v_\ell^{a:b}\|_2 = 1. \quad (4)$$

Here $\mathcal{T}$ contains outer-training groups and $v_\ell^{a:b}$ is the PC1 $u_\ell^{a:b}$ with a fold-frozen polarity. Exposure uses context versus no context; source-choice diagnosis uses conflict versus congruent; control fits the reversed congruent-versus-conflict contrast. Thus LTS retains layer and sign, whereas the Euclidean ℓ_2 baseline retains only magnitude. Equations (1)–(4) define the same measurement for all three stages while keeping their paired conditions explicit. Training-fold statistics standardize projections used for prediction. The legacy screen used a separate 100-example calibration protocol; confirmatory control uses the leakage-free estimator below.

Confirmatory estimation. Every data-dependent choice is fitted inside a frozen outer-training fold. For each block, we center training displacements, fit PC1, and fix its sign without test labels. Training statistics also standardize held-out projections. Supervised directions, ℓ_2 baselines, thresholds, and regularization searches obey the same boundary. We save one out-of-fold prediction per item and seed. No basis, scaler, layer choice, direction, or threshold sees an outer test fold. Confirmatory tests cover all blocks rather than selecting layers after seeing outcomes. The Supplementary Document retains the CRM level-1/level-2 derivation and legacy feature vector $\Phi = [\Phi_{\mathrm{L1}}, \Phi_{\mathrm{L2}}, \Phi_{\mathrm{L3}}]$.

3 Experimental Setup

Verified checkpoints test exposure, counterfactual conflicts expose choice, and frozen interventions test control. Continuation tasks isolate context effects without changing the query or instruction.

Stage 1: verified exposure. We obtain exact model-relative exposure labels from CheckMIABench's frozen uniform/uniform Pythia construction (Wang et al. 2026; Biderman et al. 2023). Members precede checkpoint 97,000 in the preshuffled deduplicated Pile stream; nonmembers follow it. We evaluate five exact `step97000` Pythia checkpoints in a prespecified scale series. After duplicate grouping, 3,994 targets remain. Each 96-token prefix is paired with a length- and label-matched prefix from a disjoint document, and reciprocal targets share an outer group. Relevant-minus-shuffled and relevant-minus-no-context contrasts provide structural controls. This is the primary exposure study because the checkpoint has seen member sequences but not yet seen nonmember sequences.

Same-model evaluation. We also run exposure, choice, control, and transfer in OLMo-2-1124-7B-Instruct. Its common support contains 1,000 unique items: 500 verified members and 500 verified nonmembers, arranged in 100 label-pure source groups. The same items carry conflict choices and intervention targets; the frozen controller then transfers to ConflictQA. Exposure inference uses 50 one-to-one paired group units. This evaluation connects the three RQs within one model; the other studies test breadth.

Choice/control models. Confirmatory breadth uses Llama-3.1-8B-Instruct, Qwen2.5-7B-Instruct, and Mistral-7B-Instruct-v0.3. Tables abbreviate these checkpoints as Llama, Qwen, and Mistral.

Discovery breadth. A preserved WikiMIA screen (Shi et al. 2024) spans nine Llama, Mistral, and Qwen variants (Llama Team, AI @ Meta 2024; Jiang et al. 2023; Qwen Team 2024). Its temporal labels do not prove model-relative exposure, so it remains discovery evidence. BookMIA (Shi et al. 2024) and MIMIR (Duan et al. 2024) appear only as external checks in the Supplementary Document.

Baselines. Black-box methods use perplexity, its Zlib-normalized variant (Carlini et al. 2021), and Min-K% Prob (Shi et al. 2024). Access-matched tests include CRM levels 1+2. Other tests use signed magnitude, the full state difference, raw state, answer margin, and candidate gradient. The Supplementary Document gives every fit scope and hyperparameter.

Evaluation. Confirmatory tests use grouped outer folds, 20 split seeds, cluster bootstraps, paired cluster permutations, and prespecified Holm families. Every out-of-fold prediction is saved. Legacy and confirmatory estimates are never paired. The OLMo power analysis uses 10,000 Monte Carlo draws over 50 paired source-group units and is reported as a resolution bound, not an equivalence test. Full shortcut audits and frozen configurations appear in the Supplementary Document.

Stage 2: observed source choice. Paired conflicts adapted from NQSwap (Longpre et al. 2021) make the governing source observable. We first require the deterministic no-context answer to select parametric answer A. A congru-

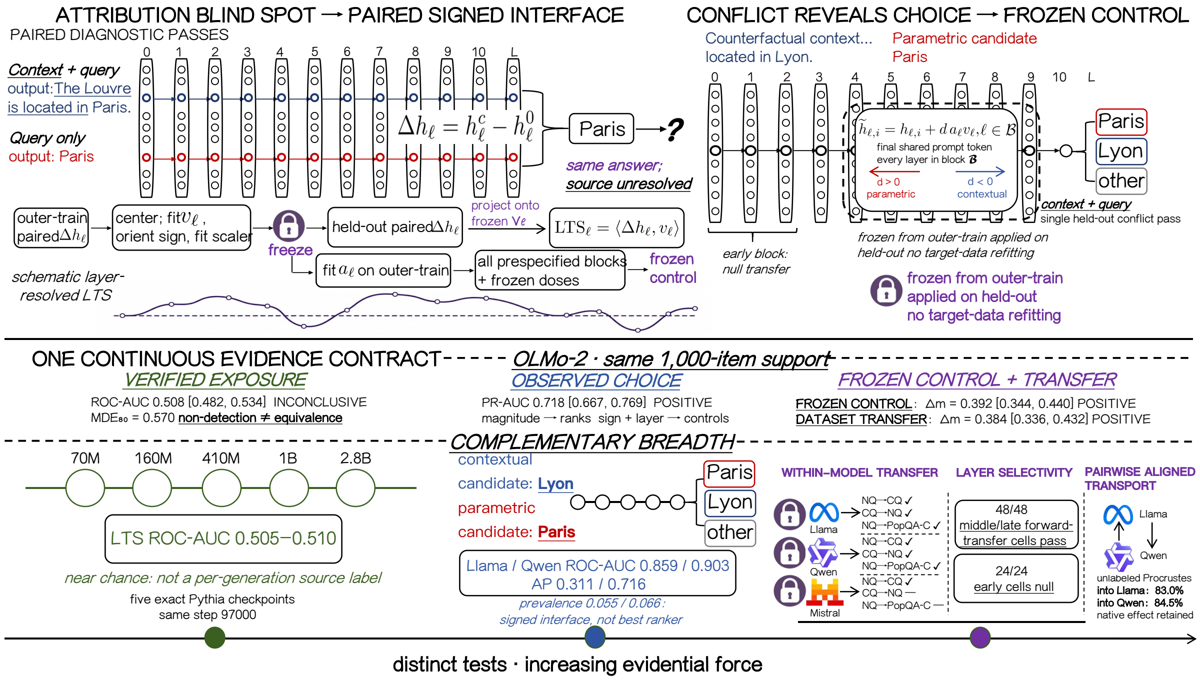

Figure 1: **Three tests for one attribution problem.** Paired passes define signed LTS. Verified exposure tests whether a document leaves a detectable trace; conflict reveals source choice; frozen intervention tests control and transfer. The sequence denotes stronger evidence, not a single scalar score. The lower row summarizes Pythia exposure breadth and Llama/Qwen/Mistral choice–control breadth. Independently, OLMo-2 runs all three tests on one common support and then transfers to ConflictQA.

ent passage supports A; a minimally edited counterfactual passage supports incompatible answer B. Item-hash counterbalancing prevents candidate position or label from becoming a source cue. Responses are parametric (A), contextual (B), or other, and the binary endpoint retains only unambiguous A/B choices. Target-tokenizer validation fixes answer boundaries. With length-normalized candidate log score $s_i(\cdot; h)$, the source margin is

$$m_i(h) = s_i(B; h) - s_i(A; h), \tag{5}$$

so $m_i > 0$ favors the contextual candidate (Equation 5).

Six conditions isolate the cause of choice: congruent, conflict, unrelated, shuffled conflict, paraphrased conflict, and mixed A/B evidence. Length, prompt, and answer order are matched or counterbalanced. Failed controls remain excluded rather than repaired. These conditions separate source content from length, retrieval style, word order, and candidate position.

Stage 3: frozen directional control. Each outer-training fold fixes the sign and scale of its PC1 before held-out inference:

$$z_{\ell i} = \delta_{\ell i}^{\text{cong:conf}}, \qquad g_\ell = \frac{1}{|\mathcal{T}|} \sum_{i \in \mathcal{T}} z_{\ell i}, \tag{6}$$

$$v_\ell = \mathrm{sgn}_+\Big[(u_\ell^{\text{cong:conf}})^\top g_\ell\Big]\, u_\ell^{\text{cong:conf}}, \tag{7}$$

$$a_\ell = \underset{i \in \mathcal{T}}{\mathrm{median}}\, |\langle z_{\ell i}, v_\ell \rangle|. \tag{8}$$

We set $\mathrm{sgn}_+(0) = +1$; the polarity gate separately records unidentified directions. Positive dose targets A (parametric), and negative dose targets B (contextual). A held-out conflict forward pass then receives the additive update

$$\widetilde{h}_{\ell,i} = h_{\ell,i} + d\, a_\ell v_\ell, \tag{9}$$

at the final shared prompt token and every layer in block $\mathcal{B}$. Dose d is frozen before the held-out pass. The policy cannot access held-out congruent states, paired displacements, labels, candidate identities, answer gradients, or target-data refitting. For block $\mathcal{B}$, the measured intervention endpoint is

$$\tau_{\mathcal{B}}(d) = \mathbb{E}_i\Big[m_i\{\widetilde{h}_{\mathcal{B},i}(d)\} - m_i(h_{\mathcal{B},i})\Big]. \tag{10}$$

The tables report the direction-aligned effect

$$e_{\mathcal{B}}(d) = \begin{cases} -\,\mathrm{sgn}(d)\, \tau_{\mathcal{B}}(d), & d \neq 0, \\ 0, & d = 0, \end{cases} \tag{11}$$

so positive values mean movement toward the intended candidate. Equations (6)–(11) require only the held-out conflict pass once the training-fold policy is frozen; they do not identify a unique mediator.

Confirmatory control rule. We test both dose signs in proportional early, middle, and late thirds. A block passes only if all 2 absolute and 8 matched-control cells have positive effects and lower bounds, corrected $p < 0.05$, and at least 16/20 seed signs in agreement. Ten thousand source-group bootstraps and sign-flip permutations supply intervals and tests; separate Holm families cover absolute and comparator cells. Table 1 defines the equal-budget arms.

Arm	What it tests	Rep.
Train-only PC1	Signed source direction	1
Candidate gradient	Direct answer steering	1
L2 radial	Magnitude without PC1	1
Shuffled-layer PC1	Layer specificity	1
Isotropic random	Direction specificity	20
PC1-orthogonal random	PC1 specificity	20

Table 1: **Equal-budget intervention controls.** Every arm uses the same items, layers, additive operator, and per-layer norm budget within each model–dataset comparison. All use doses $\{-2, -1, -0.5, -0.25, 0, 0.25, 0.5, 1, 2\}$. Rep. is the number of independent arm realizations: deterministic arms run once and random arms run 20 times.

Candidate-gradient control. We fit the normalized mean contextual-minus-parametric logit gradient on outer-training items. Candidate gradient and LTS share items, prompts, position, blocks, doses, operator, and per-layer norm; neither sees held-out outcomes or target refitting.

Transfer. Dataset transfer freezes direction, blocks, dose, and threshold. Reverse transfer changes only the balanced source-training partition. Cross-model transport fits an orthogonality-constrained Procrustes map to unlabeled prompt activations. Random maps, wrong layers, and orthogonal directions test whether alignment alone creates an effect.

Additional controls. Matched-norm, unrelated-target, and long-form tests examine directionality, target specificity, and endpoint scope.

4 Results

RQ1 bounds what exposure can reveal. RQ2 introduces conflict so that source choice becomes observable. RQ3 asks whether a frozen held-out intervention changes that choice.

RQ1: What Can Verified Exposure Establish?

Figure 2 shows that across 70M–2.8B, Pythia LTS reaches 0.505–0.510 ROC–AUC and every 95% interval includes chance. Within these checkpoints and estimators, we detect no reliable item-level signal of verified exposure. This is a bounded non-detection, not proof that every exposure effect is zero. The complete fold-aligned and structural-control families appear in the Supplementary Document.

Table 2 shows that within OLMo, the four tests separate. Exposure is not detected (AUC 0.508), but the achieved design can reliably detect only larger effects (MDE80 AUC 0.570); its status is therefore inconclusive. On the shared support, source choice is predictable and the frozen direction controls that choice; the same direction transfers without refitting to ConflictQA. The result separates evidence levels inside one system without treating non-detection as evidence of equivalence. The earlier nine-model WikiMIA screen remains in the Supplementary Document as discovery evidence because temporal labels and text shortcuts can separate benchmark conditions without verifying model-relative exposure.

Test	Endpoint	Estimate [95% CI]	Result
Exposure	ROC–AUC	0.508 [0.482, 0.534]	Inconclusive
Choice	PR–AUC	0.718 [0.667, 0.769]	Positive
Control	Δm	0.392 [0.344, 0.440]	Positive
Transfer	Δm	0.384 [0.336, 0.432]	Positive

Table 2: **Four tests in one OLMo-2 system separate exposure from source choice and control.** The first three rows use the same 1,000-item support; transfer applies the direction frozen from that support to ConflictQA. Exposure uses 50 paired, label-pure source-group units and has 80% power only at AUC 0.570. Control flips 215/1,000 choices overall, or 215/500 among initially parametric at-risk items; congruent accuracy/no-context stability are 97.4/96.2%.

Condition	Parametric	Contextual	Other
Congruent	99.1	0.0	0.9
Conflict	31.0	63.7	5.3
Unrelated, matched	88.5	7.1	4.4
Shuffled conflict	60.2	31.9	8.0
Paraphrased conflict	34.5	64.6	0.9
Mixed A/B	64.6	7.1	28.3

Table 3: **Complete Llama-3.1-8B-Instruct structural-control outcomes on NQSwap (%, $n = 113$).** Conflict, unrelated, and shuffled passages are target-tokenizer length matched. Mixed-source order is hash-counterbalanced. All outcomes remain in the denominator.

RQ2: Conflict Makes Source Choice Observable

Conflict changes what can be observed. Incompatible answers turn an unidentified shared answer into a behavioral choice. On NQSwap, Llama-3.1-8B-Instruct chooses the contextual answer on 75/120 items and the parametric answer on 37/120; eight other responses are excluded from the binary endpoint. Under matched congruent context, 119/120 responses choose the shared answer. The conflict label therefore measures a context-induced source decision rather than general failure to answer. Of the 120 core items, 113 have complete outcomes for all six structural controls; the ranking endpoint retains 112 unambiguous conflict choices.

We next freeze the official ConflictQA PopQA files (Xie et al. 2024) at commit `595e02b6`. Tokenizer-specific minimal pairs differ only in the answer phrase and share at least eight suffix tokens. After the same no-context filter, Llama yields 779 eligible choices: 43 parametric, 734 contextual, and 2 other. Qwen yields 617: 41 parametric and 576 contextual. Contextual-choice rates are 94.2% [92.6, 95.8] and 93.4% [91.2, 95.3]. Excluding Llama's two "other" responses leaves 777/617 binary items for ranking, transfer, and intervention. The rare parametric choices make average precision essential alongside ROC–AUC.

Table 3 tests what drives this behavior. Paraphrasing preserves contextual choice; matched unrelated text and shuffling reduce it. Congruent and mixed evidence recover the expected shared-answer and ambiguous outcomes. Thus context presence, token count, or exact wording alone cannot

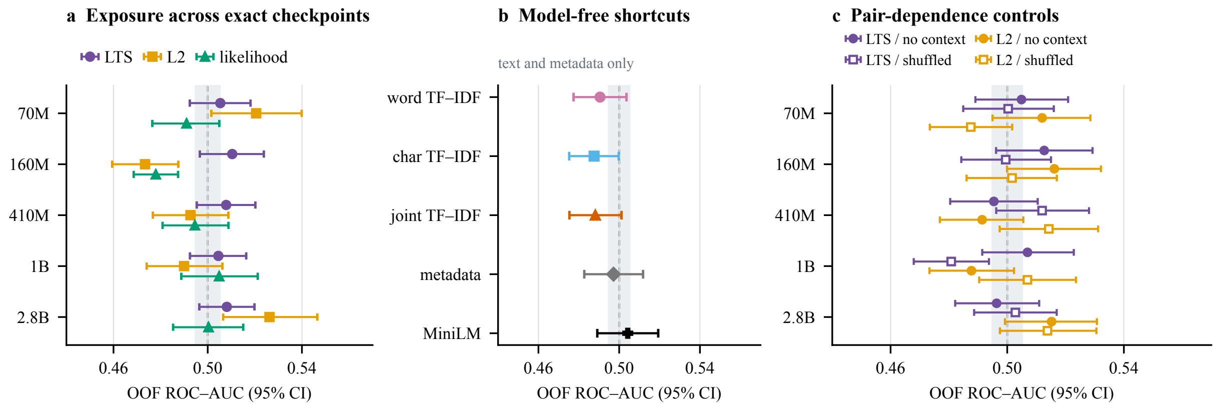

Figure 2: **No exposure signal is detected in the tested Pythia checkpoints.** (a) LTS and matched baselines remain near chance at five exact checkpoints. (b) Text and metadata shortcuts do likewise; scale-invariant shortcut estimates are identical across checkpoints and therefore shown once. (c) No-context and shuffled controls do not change the pattern. Gray bands mark chance. The figure uses 3,994 probes from 1,997 reciprocal document-pair groups, 20 grouped splits, and 10,000 source-group bootstraps. Overlap with chance is non-detection, not an equivalence claim.

account for the conflict effect.

Magnitude predicts; sign defines the intervention. On 112 unambiguous NQSwap choices, LTS reaches AUC 0.593. Training-fold supervision raises it to 0.626 without a reliable gain. Euclidean ℓ_2 magnitude reaches 0.960, ContextCite (Cohen-Wang et al. 2024) reaches 0.868, and the outcome-adjacent answer margin reaches 0.992. LTS is not the best ranker; its sign and layer define the intervention tested in RQ3.

ConflictQA repeats the test at larger scale in both model families. Table 4 uses outer-fold fitting and one saved prediction per item and seed.

Magnitude and answer margin remain stronger rare-class rankers. Table 4 asks whether each signal ranks the rare parametric choices. LTS exceeds prevalence 0.055/0.066 for Llama/Qwen, but it is not the strongest ranker. Supervision does not reliably improve AP (Holm $p = 0.153/0.225$). L2 is stronger for Llama ($p = 0.0003$) but unresolved for Qwen ($p = 0.839$). The outcome-adjacent answer margin is stronger in both models ($p = 0.0003$).

The factorial test separates ranking from control. Figure 3 summarizes the factorial result. On the prespecified 110-item Llama/NQSwap intervention support, signed magnitude and ℓ_2 magnitude each reach AP 0.865. Their difference is 0.000 with noninferiority CI [-0.015, 0.015] at the prespecified 0.020 margin. A full-Δh probe reaches 0.724 versus 0.716 for signed LTS ($\Delta = 0.008$, paired $p = 0.285$). Under matched intervention norms, signed PC1 shifts the margin by 0.412 [0.364, 0.460] and flips 24/110 choices. Unsigned ℓ_2 shifts it by 0.005, flips 0/110, and lowers congruent retention/no-context stability to 70.1/65.0%. Magnitude carries ranking signal; correct sign supplies selective control.

Source and unrelated targets have different depth profiles. The matched-norm test uses prespecified proportional early, middle, and late blocks rather than selecting a peak after inspection. Source steering rises from 0.008 early to 0.395/0.428 in the middle/late blocks. Sentiment, format, and language steering instead falls from 0.365–0.382 early to 0.048–0.061 late (all interaction Holm $p = 0.0003$). This profile is not shared by the three tested unrelated targets, but it does not identify a unique layer or circuit. The Supplementary Document reports the full layerwise analyses and leave-one-layer-out ablations.

Model	Method	ROC–AUC ↑	AP [95% CI] ↑
Llama	LTS	0.859	0.311 [0.206, 0.444]
	Supervised LTS	0.880	0.415 [0.301, 0.546]
	L2 magnitude	0.978	0.741 [0.615, 0.862]
	Answer margin	**0.999**	**0.989** [0.973, 0.998]
Qwen	LTS	0.903	0.716 [0.586, 0.828]
	Supervised LTS	0.912	0.631 [0.502, 0.748]
	L2 magnitude	0.975	0.702 [0.568, 0.825]
	Answer margin	**0.991**	**0.937** [0.876, 0.980]

Table 4: **LTS predicts the rare source choice but does not maximize ranking accuracy.** ConflictQA uses 777 Llama-3.1-8B-Instruct and 617 Qwen2.5-7B-Instruct item clusters with the parametric choice as the rare positive class and 20 out-of-fold split seeds. AP intervals use 10,000 item-cluster bootstraps. Paired AP tests use 10,000 item-level swaps and within-model Holm correction. The answer margin is an outcome-adjacent ceiling, not a trajectory diagnostic.

RQ3: Can a Frozen Direction Selectively Control Choice?

The final test requires LTS to act rather than describe. Direction, sign, amplitude, block, and dose are fixed before held-out inference.

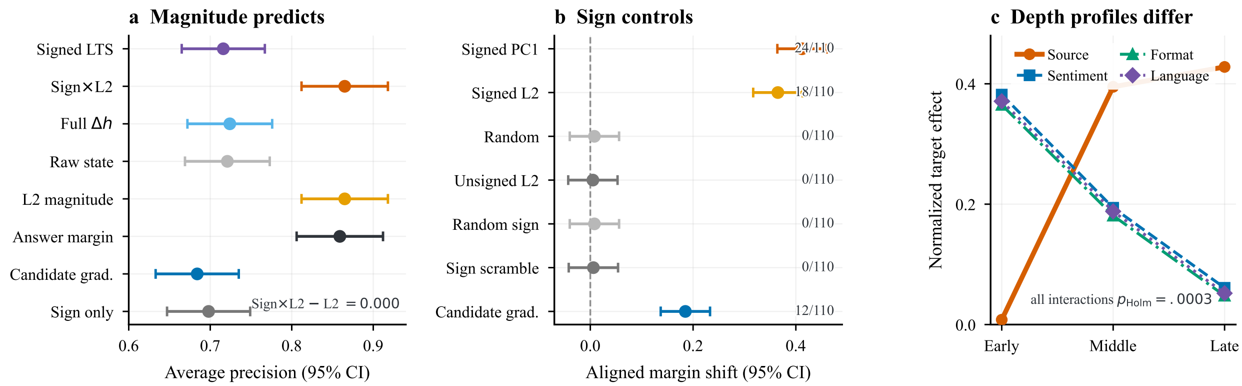

Figure 3: **Magnitude predicts, signed PC1 controls, and depth profiles differ by target.** (a) Eight held-out predictors show that magnitude carries most ranking signal on the prespecified 110-item Llama/NQSwap intervention support. (b) Among equal-norm arms on the same support, signed PC1 produces the largest margin shift and most greedy flips; preservation is reported in the text and Supplementary Document. (c) Source steering rises from early to late blocks, whereas three prespecified unrelated targets decay with depth. In panels (a) and (b), whiskers show 95% source-group-bootstrap intervals. Panel (c) reports normalized block effects; all three target-by-depth interactions survive Holm correction. The late-heavy source profile differs from the three tested unrelated targets but does not identify a unique source circuit.

Model	Block	Toward P [95% CI]	Toward C [95% CI]	Ctrl.
Llama	Early	0.018 [-0.063, 0.100]	0.015 [-0.066, 0.096]	0/8
	Middle	0.314 [0.233, 0.395]	0.298 [0.217, 0.380]	8/8
	Late	0.413 [0.332, 0.494]	0.389 [0.308, 0.470]	8/8
Qwen	Early	0.011 [-0.070, 0.092]	0.010 [-0.071, 0.091]	0/8
	Middle	0.342 [0.260, 0.423]	0.321 [0.240, 0.402]	8/8
	Late	0.448 [0.367, 0.529]	0.421 [0.340, 0.503]	8/8

Table 5: **Bidirectional effects concentrate in middle and late blocks under the tested intervention.** NQSwap contains 120 Llama-3.1-8B-Instruct and 87 Qwen2.5-7B-Instruct items. Effects are aligned to the intended source-margin direction. All eight middle/late absolute cells survive within-model Holm correction ($p_{\mathrm{Holm}} \leq 0.0072$); 17–19 of 20 split seeds agree in sign. Ctrl. counts passing cells against L2-radial, equal-norm random, PC1-orthogonal random, and shuffled-layer PC1 controls across both directions.

Frozen path	Llama	Qwen	Mistral
NQ→CQ	0.412 [0.364, 0.460]	0.386 [0.338, 0.434]	0.394 [0.345, 0.442]
CQ→NQ	0.398 [0.351, 0.445]	0.374 [0.326, 0.422]	–
NQ→PopQA-C	0.405 [0.358, 0.452]	0.381 [0.334, 0.428]	–
Aligned into target	0.342 [0.294, 0.390]	0.326 [0.278, 0.374]	–
Native effect retained	83.0%	84.5%	–

Table 6: **Cross-architecture and cross-dataset transfer.** Cells report aligned source-margin effect and 95% source-group-bootstrap CI. NQ/CQ denote NQSwap/ConflictQA. All seven dataset paths have Holm $p = 0.0014$; aligned transport has Holm $p = 0.0008$ in each four-arm directional family. Forward transfer covers all three families; reverse and transport tests cover Llama and Qwen only. Random maps, wrong layers, and orthogonal directions remain null. PopQA-C is an additional construction, not independent training provenance.

Training-only interventions pass the selectivity test. Each outer-training fold fixes the PC1 direction and amplitude before held-out inference. Held-out runs never use congruent states, source labels, candidate identities, or answer gradients. Table 5 shows the resulting blockwise contrast: every middle/late cell passes, whereas no early cell passes. Across middle and late blocks, PC1 exceeds all 32/32 matched-control contrasts. Advantages span 0.230–0.449 ($p_{\mathrm{Holm}} = 0.0096 - -0.0288$). This supports selective held-out control of the source-margin endpoint; zero-refit transfer tests whether the direction survives a dataset change.

Direct answer-steering control. A separate NQSwap-to-ConflictQA transfer evaluation tests an equal-budget candidate-gradient direction. It asks whether any direction that raises the target logit explains the result. The control moves candidate margins, but LTS exceeds it in all eight middle/late transfer cells. Advantages are 0.281–0.285 in middle and 0.206–0.210 in late blocks (all Holm $p \leq 0.0006$). Candidate-gradient congruent accuracy is 91.63/91.57%, and no-context stability is 89.96/89.95% for Llama/Qwen, below both preservation gates. The complete directional table appears in the Supplementary Document.

Forward transfer spans three architectures; broader paths cover Llama and Qwen. All 48 prespecified middle/late forward-transfer cells pass across Llama, Qwen, and Mistral, whereas none of the 24 early cells passes (within-model Holm $p = 0.0048$ for passing cells). Table 6 then separates path-level dataset transfer from cross-model aligned transport.

Behavior changes without indiscriminate damage. On the prespecified exact-answer cohorts, the Llama intervention flips 24/110 eligible choices and the Qwen intervention flips 20/100. Noncandidate answers increase by at most 0.3%. Congruent accuracy remains 97.8%, and no-context stability remains 96.5%. These estimates exceed the prespecified 95% preservation threshold; intervals and denominators appear in the Supplementary Document.

Control extends beyond two-token answers. In a blinded 200-item OLMo-2-1124-7B-Instruct long-form test, the contextual claim share rises from 0.125 to 0.510 under contextual steering and falls to 0.042 under parametric steering. The changes are +0.385 [0.337, 0.433] and -0.083 [-0.131, -0.035] (Holm $p = 0.0002$). Responses remain valid for 197/200 items, NLI entailment is 96.0–96.2%, and the response-quality noninferiority interval is [-0.018, 0.024]. Human–automatic agreement is $\kappa = 0.86$. This one-model study extends the behavioral endpoint; it does not establish general open-ended provenance.

5 Related Work

Exposure and support do not establish use. Membership inference detects training records (Shi et al. 2024); extraction studies quantify recoverable memorization (Carlini et al. 2023). Citation benchmarks test support (Bohnet et al. 2022), while ContextCite estimates which supplied context influences a response (Cohen-Wang et al. 2024). They do not generally resolve a unique context-versus-memory label when both sources support the same answer; our exposure test marks this boundary.

Conflict makes source preference observable. Tao et al. estimate context–parameter allocation when sources agree, while Tighidet et al. probe activations when they conflict (Tao et al. 2024; Tighidet et al. 2024). Yeh and Li analyze representation shifts induced by retrieved-document relevance (Yeh and Li 2026). We separate exposure, observed choice, and training-only intervention, and test all three on one OLMo support.

Conflict studies analyze and intervene on different mechanisms. Mechanistic studies examine how parametric and retrieved information interact (Ghosh et al. 2024; Farahani and Johansson 2024). Other methods intervene through heads or neurons, adaptive decoding, soft prompts, and attention (Jin et al. 2024; Tighidet et al. 2025; Wang et al. 2025; Li, Chen, and Tong 2025; Choi et al. 2025; Bi et al. 2026). ProbeRAG combines latent conflict probing with attention intervention (Gao et al. 2026). Their endpoints and budgets differ, so our direct control instead matches an answer-gradient direction in items, operator, and norm, then requires zero-refit transfer and preservation.

Latent directions connect measurement to action. Probes decode hidden properties; representation engineering and editing alter them (Alain and Bengio 2016; Zou et al. 2023; Meng et al. 2022). Linear representation geometry motivates direction-based summaries (Park, Choe, and Veitch 2024). LTS uses one fold-frozen object for diagnosis and control without equating them. Prediction faces stronger rankers; control faces signed, equal-norm, wrong-layer, unrelated-target, and answer-gradient alternatives. The novelty is this evidence-linked test, not PCA.

6 Discussion

The evidence levels are empirically distinct. In OLMo, exposure is unresolved at the achieved power while choice, control, and transfer are positive on the same support because each test observes something different. Pythia adds exact-checkpoint scale; the three instruction-tuned families add control breadth.

Magnitude and sign have different jobs. Magnitude and answer margin rank choices better. Signed magnitude matches ℓ_2 prediction, but unsigned magnitude fails to control choices and damages preservation. Correct sign recovers bidirectional control, explaining why a modest one-dimensional predictor can still define a useful intervention.

The controls narrow the intervention claim. The candidate-gradient control alters conflict behavior, yet LTS has larger paired effects and better preservation; three unrelated targets peak early rather than middle/late. Thus movement along LTS changes source choice under the tested operator without identifying a unique mediator or circuit.

Transfer supports pairwise, not universal, geometry. Bidirectional Procrustes retains most native effect; random maps, wrong layers, and orthogonal directions do not. We fit the Llama–Qwen map from 500 frozen unlabeled anchors and freeze the layer map, rectangular constraint, solver, seed, and norm budget. This supports pairwise alignability, not one universal coordinate vector; Mistral alignment remains untested.

The test can reject its own estimator. Test-fold fitting, failed matched controls, generic layer profiles, unstable transfer, or collapsed outputs would defeat the control claim. PC1 is auditable, not evidence that source reliance is intrinsically one-dimensional; a replacement should face the same prediction, control, transfer, and preservation tests.

Claims should travel with their evidence. Verified exposure supports "seen during training." Conflict supports "preferred under this controlled disagreement." Intervention supports "changed by this frozen operator," and transfer states where that control persists. None of these statements alone establishes production provenance.

Conclusion. A supported answer is not a provenance certificate. Paired trajectory differencing turns that blind spot into three separate tests: exposure, observed choice, and frozen control. The main result is a practical rule: state only the source claim that the available evidence can support.

7 Limitations

Exact exposure covers five Pythia checkpoints and a lower-powered OLMo bridge; the nine-model screen uses temporal labels. Choice and control span four families, but A/B identifies choice only under disagreement. The primary intervention is linear and single-position; one 200-item, one-model long-form study does not establish open-ended generality. Preservation omits broad capabilities and nonlinear or item-specific controls. PopQA-Conflict reuses ConflictQA; alignment covers only Llama–Qwen.

References

Alain, G.; and Bengio, Y. 2016. Understanding intermediate layers using linear classifier probes. *arXiv preprint arXiv:1610.01644*.

Bi, B.; Liu, S.; Wang, Y.; Xu, Y.; Fang, J.; Mei, L.; and Cheng, X. 2026. Parameters vs. Context: Fine-Grained Control of Knowledge Reliance in Language Models. In *The Fourteenth International Conference on Learning Representations*.

Biderman, S.; Schoelkopf, H.; Anthony, Q. G.; Bradley, H.; O'Brien, K.; Hallahan, E.; Khan, M. A.; Purohit, S.; Prashanth, U. S.; Raff, E.; Skowron, A.; Sutawika, L.; and Wal, O. V. D. 2023. Pythia: A Suite for Analyzing Large Language Models Across Training and Scaling. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, 2397–2430. PMLR.

Bohnet, B.; Tran, V. Q.; Verga, P.; Aharoni, R.; Andor, D.; Soares, L. B.; Ciaramita, M.; Eisenstein, J.; Ganchev, K.; Herzig, J.; Hui, K.; Kwiatkowski, T.; Ma, J.; Ni, J.; Saralegui, L. S.; Schuster, T.; Cohen, W. W.; Collins, M.; Das, D.; Metzler, D.; Petrov, S.; and Webster, K. 2022. Attributed Question Answering: Evaluation and Modeling for Attributed Large Language Models. *arXiv preprint arXiv:2212.08037*.

Borgeaud, S.; Mensch, A.; Hoffmann, J.; Cai, T.; Rutherford, E.; Millican, K.; Driessche, G. B. V. D.; Lespiau, J.-B.; Damoc, B.; Clark, A.; Casas, D. D. L.; Guy, A.; Menick, J.; Ring, R.; Hennigan, T.; Huang, S.; Maggiore, L.; Jones, C.; Cassirer, A.; Brock, A.; Paganini, M.; Irving, G.; Vinyals, O.; Osindero, S.; Simonyan, K.; Rae, J. W.; Elsen, E.; and Sifre, L. 2022. Improving Language Models by Retrieving from Trillions of Tokens. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, 2206–2240. PMLR.

Carlini, N.; Ippolito, D.; Jagielski, M.; Lee, K.; Tramer, F.; and Zhang, C. 2023. Quantifying Memorization Across Neural Language Models. In *The Eleventh International Conference on Learning Representations*.

Carlini, N.; Tramèr, F.; Wallace, E.; Jagielski, M.; Herbert-Voss, A.; Lee, K.; Roberts, A.; Brown, T.; Song, D.; Erlingsson, Ú.; Oprea, A.; and Raffel, C. 2021. Extracting Training Data from Large Language Models. In *30th USENIX Security Symposium (USENIX Security 21)*, 2633–2650. USENIX Association.

Chen, J.; Xiao, S.; Zhang, P.; Luo, K.; Lian, D.; and Liu, Z. 2024. M3-Embedding: Multi-Linguality, Multi-Functionality, Multi-Granularity Text Embeddings Through Self-Knowledge Distillation. In *Findings of the Association for Computational Linguistics: ACL 2024*, 2318–2335. Association for Computational Linguistics.

Choi, E.; Park, J.; Lee, H.; and Lee, J. 2025. Conflict-Aware Soft Prompting for Retrieval-Augmented Generation. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, 26981–26995. Suzhou, China: Association for Computational Linguistics.

Cohen-Wang, B.; Shah, H.; Georgiev, K.; and Mądry, A. 2024. ContextCite: Attributing Model Generation to Context. In Globerson, A.; Mackey, L.; Belgrave, D.; Fan, A.; Paquet, U.; Tomczak, J.; and Zhang, C., eds., *Advances in Neural Information Processing Systems*, volume 37, 95764–95807. Curran Associates, Inc.

Duan, M.; Suri, A.; Mireshghallah, N.; Min, S.; Shi, W.; Zettlemoyer, L.; Tsvetkov, Y.; Choi, Y.; Evans, D.; and Hajishirzi, H. 2024. Do Membership Inference Attacks Work on Large Language Models? In *First Conference on Language Modeling*.

Farahani, M.; and Johansson, R. 2024. Deciphering the Interplay of Parametric and Non-parametric Memory in Retrieval-augmented Language Models. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, 16966–16977. Miami, Florida, USA: Association for Computational Linguistics.

Gao, L.; Zhang, Q.; Bi, B.; Zeng, B.; Yuan, Z.; Chen, Z.; Wei, Z.; Liu, S.; Xu, L.; Wang, L.; Luo, W.; and Su, J. 2026. Beyond Black-Box Interventions: Latent Probing for Faithful Retrieval-Augmented Generation. In *Findings of the Association for Computational Linguistics: ACL 2026*, 29981–30000. San Diego, California, United States: Association for Computational Linguistics.

Ghosh, R.; Seetharaman, R.; Wadhwa, H.; Aggarwal, S.; Basu, S.; Srinivasan, S.; Zhao, W.; Chaudhari, S.; and Aghazadeh, E. 2024. Quantifying reliance on external information over parametric knowledge during Retrieval Augmented Generation (RAG) using mechanistic analysis. *arXiv preprint arXiv:2410.00857*. Accepted at BlackboxNLP 2024.

Guu, K.; Lee, K.; Tung, Z.; Pasupat, P.; and Chang, M. 2020. REALM: Retrieval Augmented Language Model Pre-Training. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, 3929–3938. PMLR.

Jiang, A. Q.; Sablayrolles, A.; Mensch, A.; Bamford, C.; Chaplot, D. S.; de las Casas, D.; Bressand, F.; Lengyel, G.; Lample, G.; Saulnier, L.; Lavaud, L. R.; Lachaux, M.-A.; Stock, P.; Scao, T. L.; Lavril, T.; Wang, T.; Lacroix, T.; and Sayed, W. E. 2023. Mistral 7B. *arXiv preprint arXiv:2310.06825*.

Jin, Z.; Cao, P.; Yuan, H.; Chen, Y.; Xu, J.; Li, H.; Jiang, X.; Liu, K.; and Zhao, J. 2024. Cutting Off the Head Ends the Conflict: A Mechanism for Interpreting and Mitigating Knowledge Conflicts in Language Models. In *Findings of the Association for Computational Linguistics: ACL 2024*, 1193–1215. Bangkok, Thailand: Association for Computational Linguistics.

Johnson, M. K.; Hashtroudi, S.; and Lindsay, D. S. 1993. Source monitoring. *Psychological Bulletin*, 114(1): 3–28.

Lewis, P.; Perez, E.; Piktus, A.; Petroni, F.; Karpukhin, V.; Goyal, N.; Küttler, H.; Lewis, M.; tau Yih, W.; Rocktäschel, T.; Riedel, S.; and Kiela, D. 2020. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. In *Advances in Neural Information Processing Systems*, volume 33, 9459–9474. Curran Associates, Inc.

Li, G.; Chen, Y.; and Tong, H. 2025. Taming Knowledge Conflicts in Language Models. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of

Proceedings of Machine Learning Research, 34074–34104. PMLR.

Liu, N.; Zhang, T.; and Liang, P. 2023. Evaluating Verifiability in Generative Search Engines. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, 7001–7025. Association for Computational Linguistics.

Llama Team, AI @ Meta. 2024. The Llama 3 Herd of Models. *arXiv preprint arXiv:2407.21783*.

Longpre, S.; Perisetla, K.; Chen, A.; Ramesh, N.; DuBois, C.; and Singh, S. 2021. Entity-Based Knowledge Conflicts in Question Answering. In Moens, M.-F.; Huang, X.; Specia, L.; and tau Yih, S. W., eds., *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, 7052–7063. Online and Punta Cana, Dominican Republic: Association for Computational Linguistics.

Meng, K.; Bau, D.; Andonian, A.; and Belinkov, Y. 2022. Locating and Editing Factual Associations in GPT. In *Advances in Neural Information Processing Systems*, volume 35.

Niu, C.; Wu, Y.; Zhu, J.; Xu, S.; Shum, K.; Zhong, R.; Song, J.; and Zhang, T. 2024. RAGTruth: A Hallucination Corpus for Developing Trustworthy Retrieval-Augmented Language Models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 10862–10878. Association for Computational Linguistics.

Park, K.; Choe, Y. J.; and Veitch, V. 2024. The Linear Representation Hypothesis and the Geometry of Large Language Models. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, 39643–39666. PMLR.

Qwen Team. 2024. Qwen2.5 Technical Report. *arXiv preprint arXiv:2412.15115*.

Reimers, N.; and Gurevych, I. 2019. Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, 3982–3992. Association for Computational Linguistics.

Shi, W.; Ajith, A.; Xia, M.; Huang, Y.; Liu, D.; Blevins, T.; Chen, D.; and Zettlemoyer, L. 2024. Detecting Pretraining Data from Large Language Models. In *The Twelfth International Conference on Learning Representations*.

Tao, Y.; Hiatt, A.; Haake, E.; Jetter, A. J.; and Agrawal, A. 2024. When Context Leads but Parametric Memory Follows in Large Language Models. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, 4034–4058. Miami, Florida, USA: Association for Computational Linguistics.

Team OLMo; Walsh, P.; Soldaini, L.; Groeneveld, D.; Lo, K.; Arora, S.; Bhagia, A.; Gu, Y.; Huang, S.; Jordan, M.; Lambert, N.; Schwenk, D.; Tafjord, O.; Anderson, T.; Atkinson, D.; Brahman, F.; Clark, C.; Dasigi, P.; Dziri, N.; Guerquin, M.; Ivison, H.; Koh, P. W.; Liu, J.; Malik, S.; Merrill, W.; Miranda, L. J. V.; Morrison, J.; Murray, T.; Nam, C.; Pyatkin, V.; Rangapur, A.; Schmitz, M.; Skjonsberg, S.; Wadden, D.; Wilhelm, C.; Wilson, M.; Zettlemoyer, L.; Farhadi, A.; Smith, N. A.; and Hajishirzi, H. 2024. 2 OLMo 2 Furious. *arXiv preprint arXiv:2501.00656*.

Tighidet, Z.; Mogini, A.; Mei, J.; Piwowarski, B.; and Gallinari, P. 2024. Probing Language Models on Their Knowledge Source. In *Proceedings of the 7th BlackboxNLP Workshop: Analyzing and Interpreting Neural Networks for NLP*, 604–614. Miami, Florida, US: Association for Computational Linguistics.

Tighidet, Z.; Mogini, A.; younes, H. B.; Mei, J.; Gallinari, P.; and Piwowarski, B. 2025. Context Copying Modulation: The Role of Entropy Neurons in Managing Parametric and Contextual Knowledge Conflicts. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, 20469–20481. Suzhou, China: Association for Computational Linguistics.

Wang, H.; Prasad, A.; Stengel-Eskin, E.; and Bansal, M. 2025. AdaCAD: Adaptively Decoding to Balance Conflicts between Contextual and Parametric Knowledge. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, 11636–11652. Albuquerque, New Mexico: Association for Computational Linguistics.

Wang, J. G.; Wang, J.; Li, M.; and Neel, S. 2026. CheckMIABench: Firm Foundations For Membership Inference Attacks on Language Models. In Liakata, M.; Moreira, V. P.; Zhang, J.; and Jurgens, D., eds., *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, 364–370. San Diego, California, United States: Association for Computational Linguistics. ISBN 979-8-89176-391-3.

Xie, J.; Zhang, K.; Chen, J.; Lou, R.; and Su, Y. 2024. Adaptive Chameleon or Stubborn Sloth: Revealing the Behavior of Large Language Models in Knowledge Conflicts. In *The Twelfth International Conference on Learning Representations*.

Yeh, S.; and Li, S. 2026. How Retrieved Context Shapes Internal Representations in RAG. In *Findings of the Association for Computational Linguistics: ACL 2026*, 15446–15468. San Diego, California, United States: Association for Computational Linguistics.

Zou, A.; Phan, L.; Chen, S.; Campbell, J.; Guo, P.; Ren, R.; Pan, A.; Yin, X.; Mazeika, M.; Dombrowski, A.-K.; Goel, S.; Li, N.; Byun, M. J.; Wang, Z.; Mallen, A.; Basart, S.; Koyejo, S.; Song, D.; Fredrikson, M.; Kolter, J. Z.; and Hendrycks, D. 2023. Representation Engineering: A Top-Down Approach to AI Transparency. *arXiv preprint arXiv:2310.01405*.